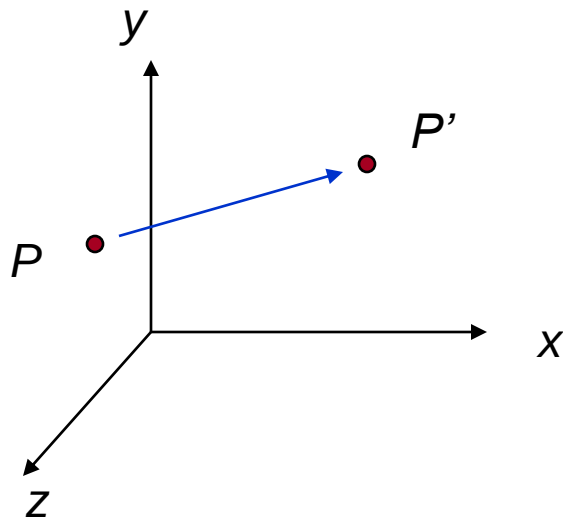


# **3D Transformations**

# Translation

---



$$x' = x + tx$$

$$y' = y + ty$$

$$z' = z + tz$$

$$P = \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

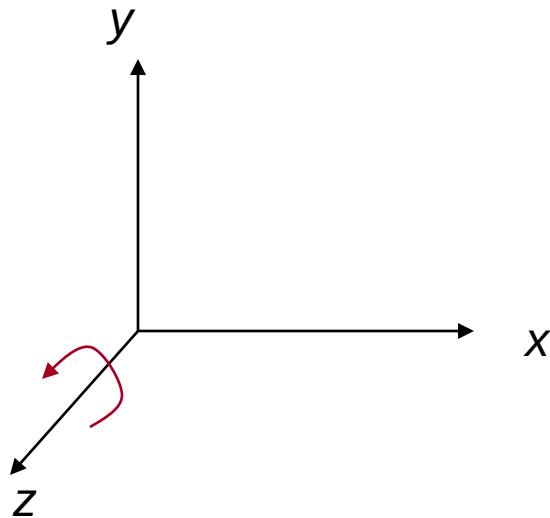
$$P' = \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix}$$

$$T = \begin{pmatrix} 1 & 0 & 0 & tx \\ 0 & 1 & 0 & ty \\ 0 & 0 & 1 & tz \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$P' = T \cdot P$$

# Rotation

---



*z-axis rotation*

$$x' = x.\cos\theta - y.\sin\theta$$

$$y' = x.\sin\theta + y.\cos\theta$$

$$z' = z$$

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$P' = R . P$$

# Rotation

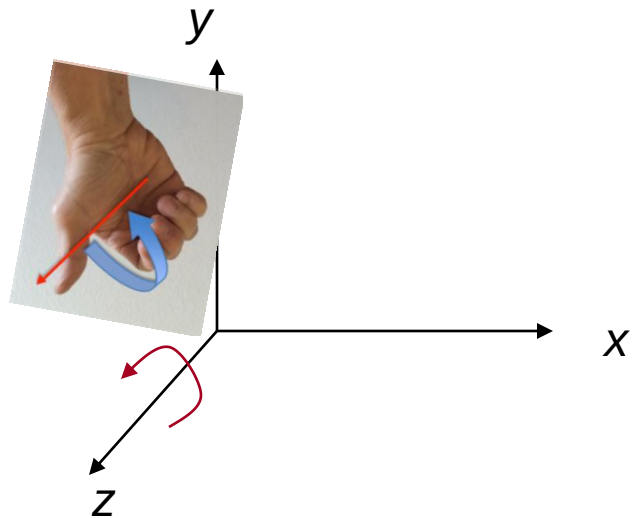
---

*z-axis rotation*

$$x' = x.\cos\theta - y.\sin\theta$$

$$y' = x.\sin\theta + y.\cos\theta$$

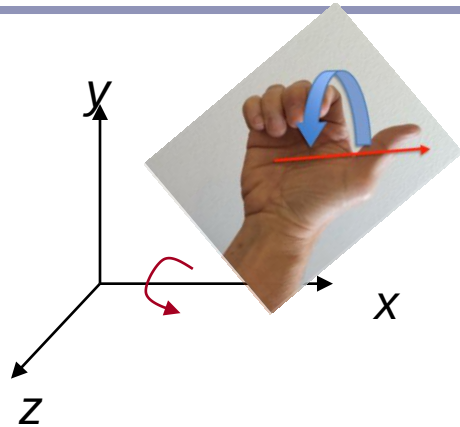
$$z' = z$$



$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$P' = R . P$$

# Rotation



## *x-axis rotation*

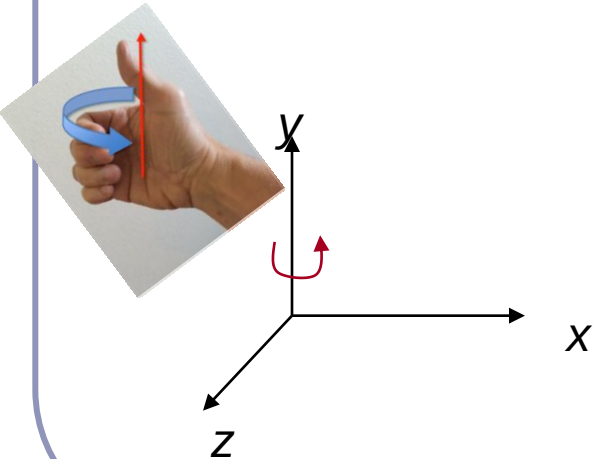
*replace  $x \rightarrow y \rightarrow z \rightarrow x$  in z-axis rotation*

$$y' = y.\cos\theta - z.\sin\theta$$

$$z' = y.\sin\theta + z.\cos\theta$$

$$x' = x$$

$$R = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



## *y-axis rotation*

*replace  $x \rightarrow y \rightarrow z \rightarrow x$  in x-axis rotation*

$$z' = z.\cos\theta - x.\sin\theta$$

$$x' = z.\sin\theta + x.\cos\theta$$

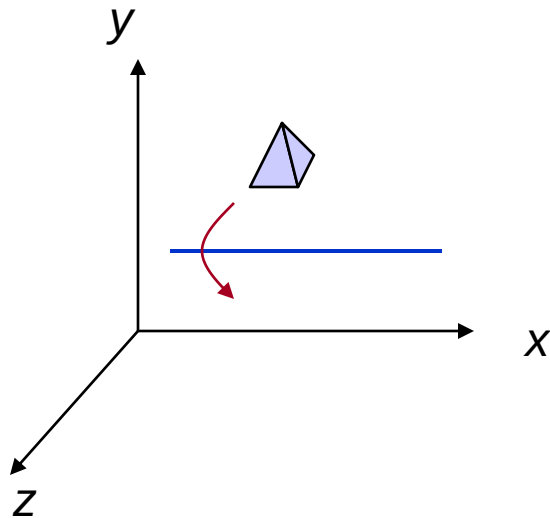
$$y' = y$$

$$R = \begin{pmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

# Rotation

---

*Rotation about an arbitrary  
axis parallel to a  
coordinate axis*

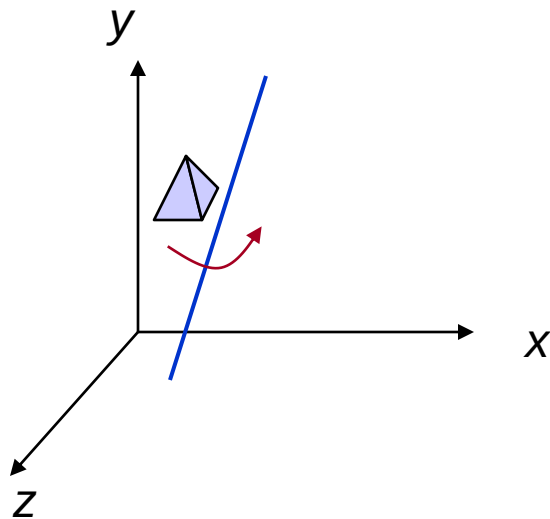


$$P' = T^{-1} \cdot Rx(\theta) \cdot T \cdot P$$

# Rotation

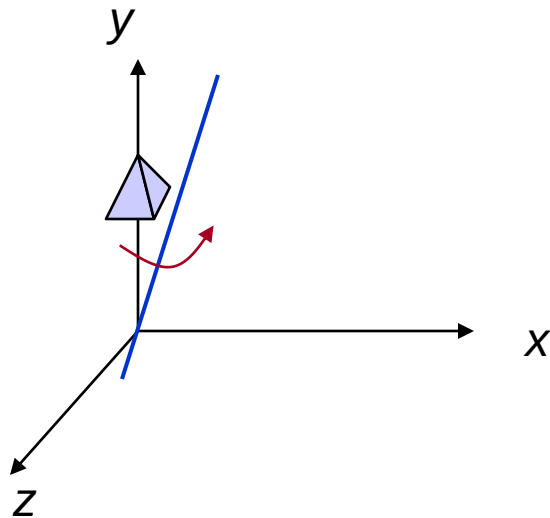
---

*Rotation about an arbitrary  
axis NOT parallel to a  
coordinate axis*



# Rotation

---

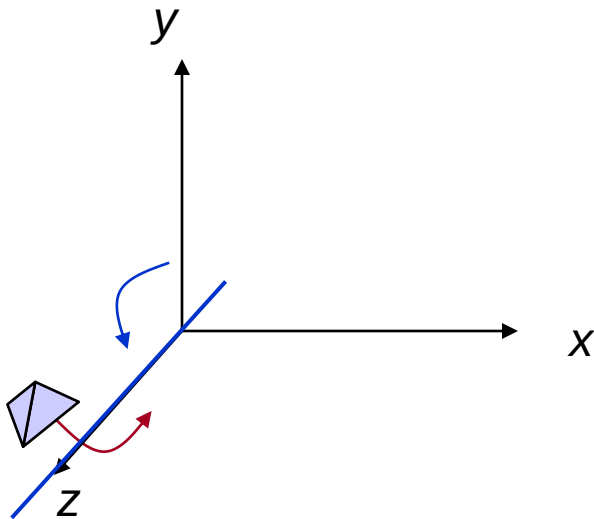


***Rotation about an arbitrary axis NOT parallel to a coordinate axis***

- 1. Translate so that rotation axis passes through the origin***

# Rotation

---

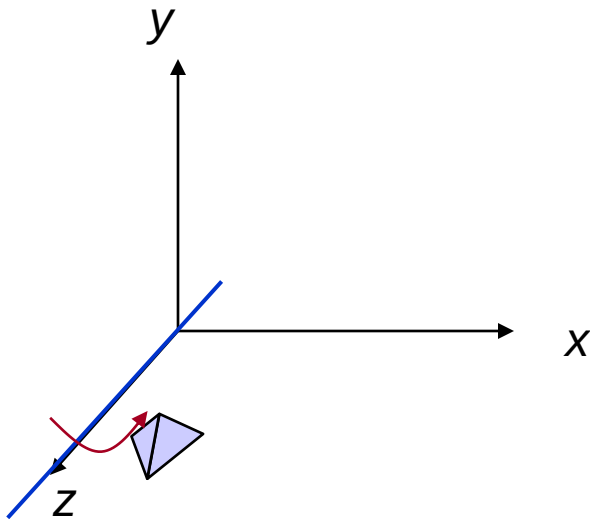


***Rotation about an arbitrary axis NOT parallel to a coordinate axis***

- 1. Translate so that rotation axis passes through the origin***
- 2. Rotate so that rotation axis coincides with a coordinate axis***

# Rotation

---

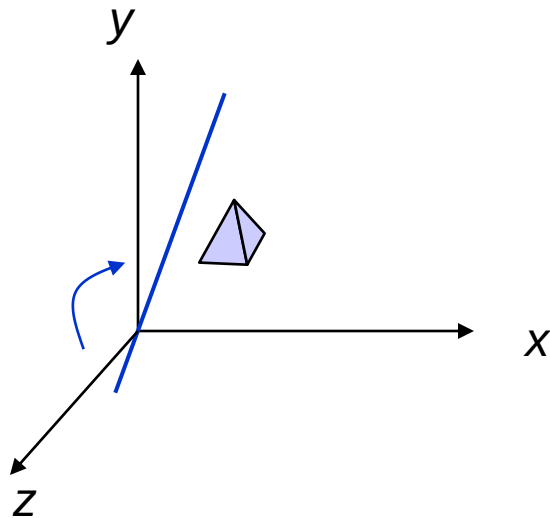


***Rotation about an arbitrary axis NOT parallel to a coordinate axis***

- 1. Translate so that rotation axis passes through the origin***
- 2. Rotate so that rotation axis coincides with a coordinate axis***
- 3. Rotate***

# Rotation

---

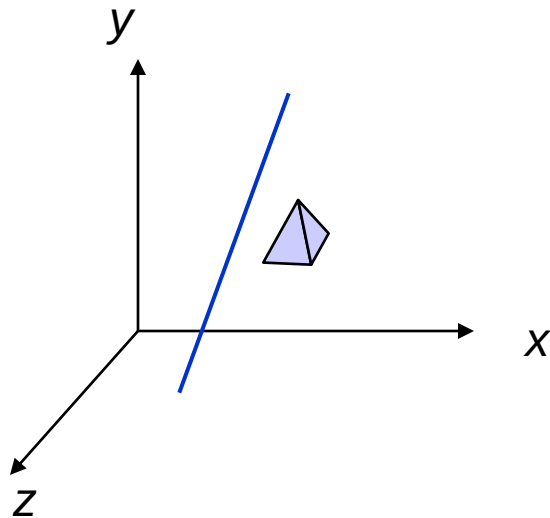


***Rotation about an arbitrary axis NOT parallel to a coordinate axis***

- 1. Translate so that rotation axis passes through the origin***
- 2. Rotate so that rotation axis coincides with a coordinate axis***
- 3. Rotate***
- 4. Inverse rotation***

# Rotation

---

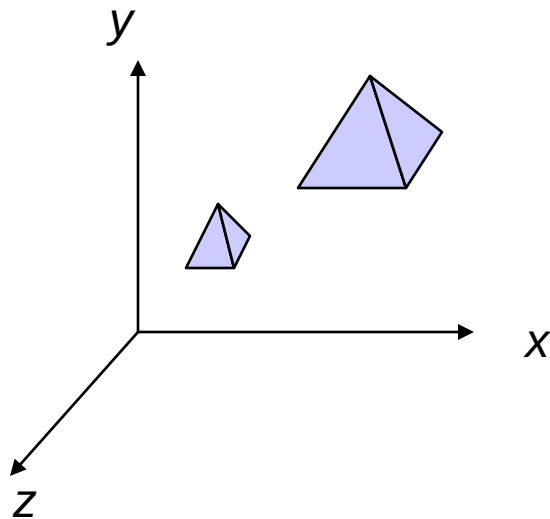


***Rotation about an arbitrary axis NOT parallel to a coordinate axis***

- 1. Translate so that rotation axis passes through the origin***
- 2. Rotate so that rotation axis coincides with a coordinate axis***
- 3. Rotate***
- 4. Inverse rotation***
- 5. Inverse translation***

# Scaling

---



$$S = \begin{pmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

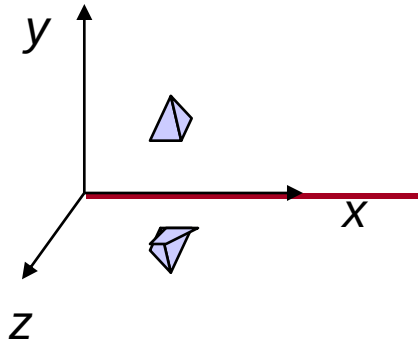
$$P' = S \cdot P$$

*wrt a fixed point (xf, yf, zf) :*

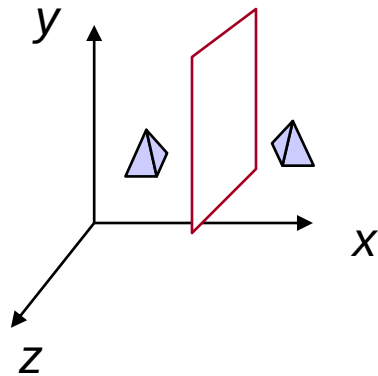
$$T(xf, yf, zf) \cdot S(s_x, s_y, s_z) \cdot T(-xf, -yf, -zf)$$

# Reflection

---



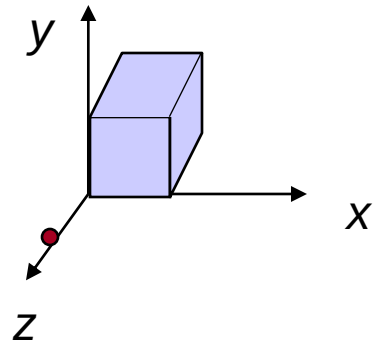
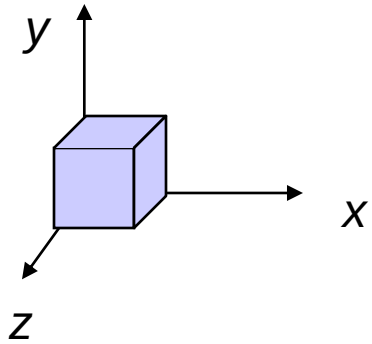
***$180^\circ$  rotation about  $x$ -axis***



***a combination of translation  
and rotation***

# Shear

---



$$\begin{pmatrix} 1 & 0 & shx & -shx.zref \\ 0 & 1 & shy & -shy.zref \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

# OpenGL

---

- **glTranslate\*(tx, ty, tz)**
  - f (float)
  - d (double)
- **glRotate\* (theta, vx, vy, vz)**
  - (vx, vy, vz) vector defines the orientation of the rotation axis that passes through the coordinate origin
- **glScale\*(sx, sy, sz)**